P4 - Final assignment

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Introduction, explaining the choice of region: Lüderitz, Namibia

I chose the Lüderitz region in Namibia because it features a coastal upwelling zone with high chlorophyll concentrations near the shore, contrasting with lower levels farther offshore in the oligotrophic waters. This makes it easy to observe variations in the mean annual chlorophyll and seasonal productivity. The Lüderitz area is part of the Benguela Current system, one of the most productive eastern boundary upwelling systems (Moloto et al., 2023). Lüderitz experiences upwelling of cold, nutrient-rich waters along the continental shelf, which boosts phytoplankton productivity, as shown by higher chlorophyll levels near Lüderitz.

Furthermore, Lüderitz's steep continental shelf makes bathymetry easy to see and map. The bathymetry around Lüderitz greatly influences ocean circulation and upwelling intensity, which in turn makes the chlorophyll-a gradient more noticeable, as nutrient supply and biological productivity vary widely in this area (Moloto et al., 2023). I selected a regional scale of about 200-300 km to capture detailed coastal processes while maintaining the high spatial resolution of the satellite-derived chlorophyll data.

Dataset 1: GMRT Bathymetry

Description:

Bathymetry data for the Lüderitz coastal region were obtained from the Global Multi-Resolution Topography Data Synthesis (GMRT). GMRT is a gridded digital elevation model that combines seafloor bathymetry and land topography into a single dataset. The GMRT data were downloaded as a NetCDF file and subsetted to the Lüderitz coastal region for bathymetry plotting.

Format: NetCDF (.nc) gridded dataset.

Temporal scale: Static dataset without a time dimension.

Spatial coverage: The subset for Lüderitz spans approximately 10.5 °E to 16.0 °E and 25.5 ˚S to 29.5 °S.

Spatial resolution:

The GMRT bathymetry was downloaded at a medium grid resolution in NetCDF format, following the Climate and Forecast convention option (Coards/CF Compliant NetCDF, GMT id: nd). This resolution captures the main bathymetric features of the Lüderitz coastal region, and aligns well with the approximately 4 km resolution of the chlorophyll-a dataset.

Coordinates and data variables:

- lat = latitude
- lon = longitude
- altitude = elevation/depth

Units: Metres (m)

Source: Global Multi-Resolution Topography Data Synthesis (GMRT).

Projection/coordinate system: The file metadata indicates a "Cylindrical Equidistant" projection for the longitude-latitude grid.

Dataset 2: ESA-CCI Ocean Colour Chlorophyll-a Climatology

Description:

Chlorophyll-a data for the Lüderitz coastal region were obtained from the ESA-CCI Ocean Colour climatology. The ESA-CCI Ocean Colour climatology is an extensive multi-sensor satellite dataset, focused on climate research, which also includes data on chlorophyll-a. The global monthly climatology was subset to the Lüderitz coastal region and used to analyse the spatial and seasonal variations of surface chlorophyll-a levels. Since chlorophyll-a is a common satellite proxy for phytoplankton biomass, the dataset reflects the abundance of phytoplankton.

Format: NetCDF (.nc) gridded dataset.

Temporal scale: Monthly climatology with 12 monthly time steps.

Spatial coverage:

The original dataset had global coverage, but was subset for Lüderitz to span roughly 10.5 °E to 16.0 °E and 25.5 °S to 29.5 °S.

Spatial resolution: Approximately ~ 4 km resolution.

Coordinates and data variables:

- time = month
- lat = latitude
- lon = longitude
- chlor_a = chlorophyll-a concentration in seawater

Units: Milligrams per cubic meter ($mg\ \ m^{-3}$)

Source: ESA Ocean Colour Climate Change Initiative (Ocean Colour CCI)

Projection/ coordinate system: Geographic longitude and latitude grid.

Figures:

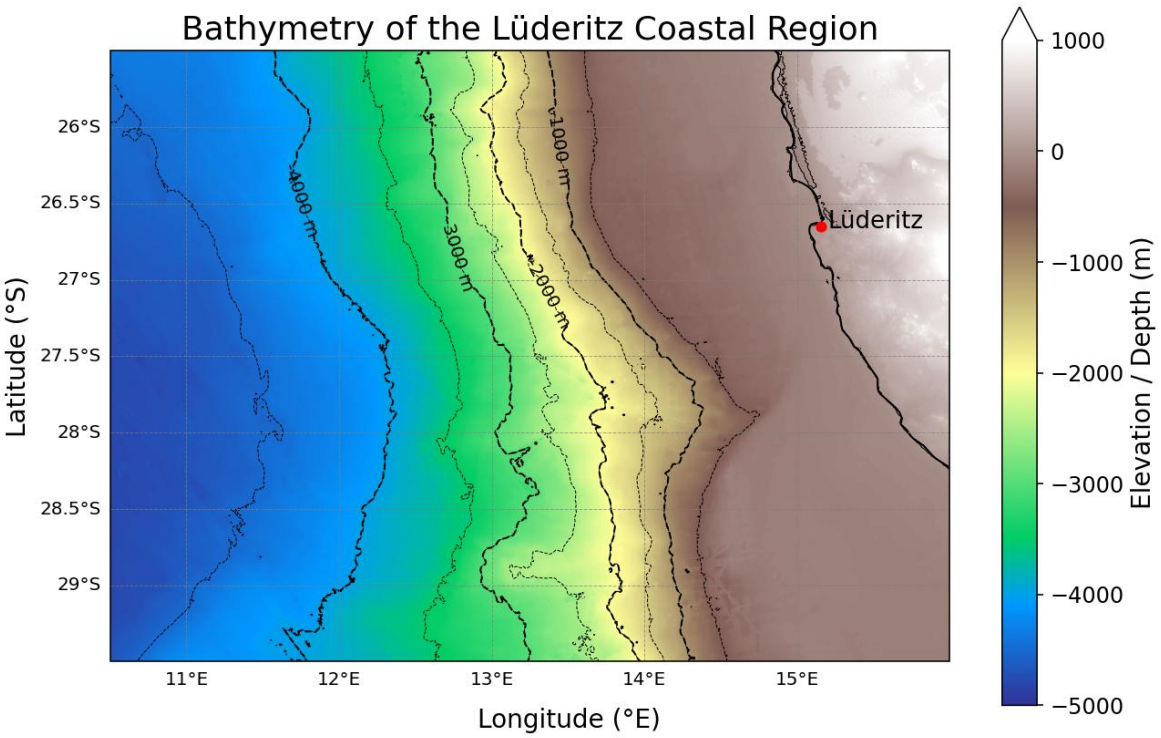

Figure 1: Bathymetry from the GMRT dataset for the Lüderitz region shows the main depth contours, with Lüderitz marked on the coast.

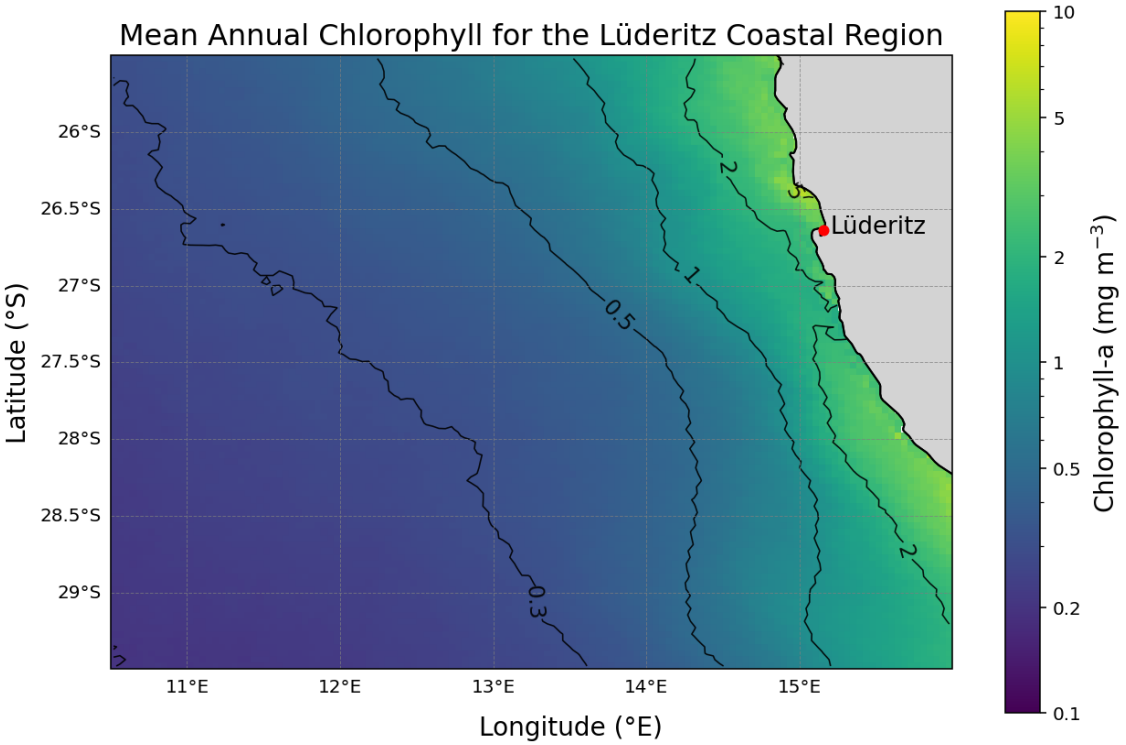

Figure 2: Annual mean chlorophyll-a concentration, derived from the ESA-CCI ocean colour climatology, plotted on a logarithmic colour scale to better highlight the strong chlorophyll-a gradient from coast to open ocean, with the highest concentrations near the coast around Lüderitz.

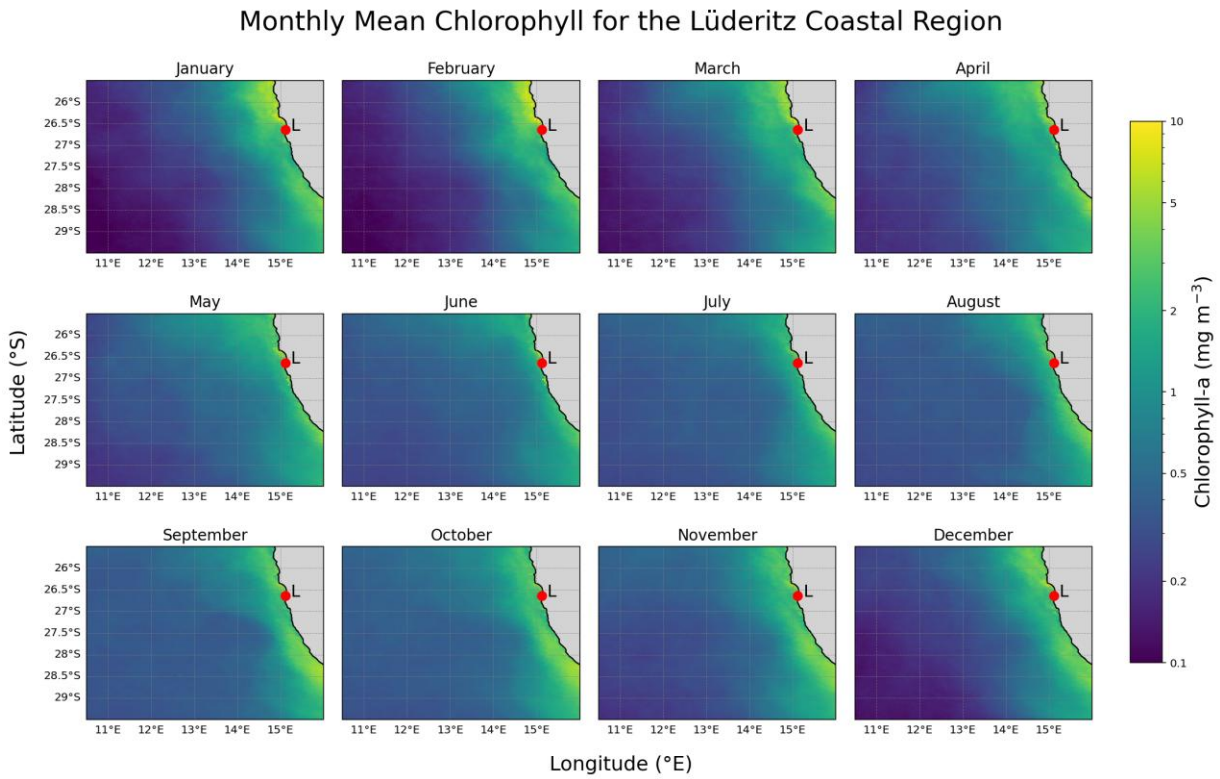

Figure 3: Faceted, monthly climatological chlorophyll-a maps showing the seasonal cycle of surface chlorophyll in the Lüderitz coastal region. The red dot and "L" denote the location of Lüderitz, and the colour scale is logarithmic to better depict spatial variability in the chlorophyll concentration.

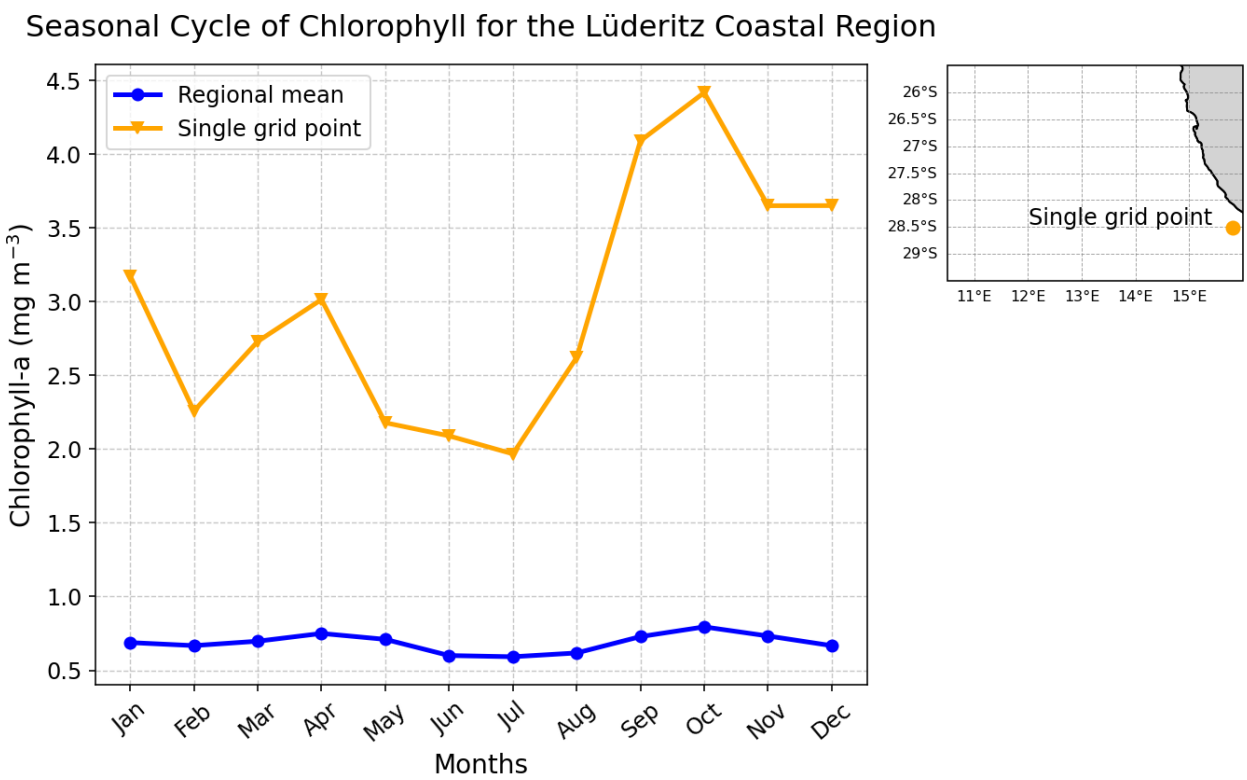

Figure 4: Comparison of the average regional seasonal cycle of chlorophyll-a with the seasonal pattern at a specific coastal grid point near Lüderitz. The map shows the location of that grid point.

References:

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